



Mustang Math Tournament 2025



Focused Filly Stallion Round



Basic Format

- This round contains 20 problems to be solved in 60 minutes.
- The problems are divided into four sets (Algebra, Geometry, Number Theory, Combinatorics) with five problems each.
- The more problems you solve in a single set, the more points you will receive per problem. You receive 4 points for the first problem solved, 5 points for the next problem you solve, and so on up to 8 points if you solve the last problem.
- For instance, if you solve two problems of the Algebra set, you will receive 9 points total for that set (regardless of which two you solved). This makes for a maximum of 30 points per set.
- **Do not** write below the provided answer blank inside each space on the answer sheet (the space is for grading purposes).



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**WE WILL NOT GRADE ANY WORK ON THIS
PAGE.**



Algebra (A)

- [A1]. Jeremy takes two different digits from the number 173205 and multiplies them. What is the largest possible result?
- [A2]. The set of positive integers $\{4, 4, a, b, c\}$ has no unique mode, and the mean and median are equal. What is the least possible value of $a + b + c$?
- [A3]. Two perpendicular lines are drawn in the coordinate plane, intersecting at the point $(20, 25)$. A third line with slope 13 is drawn, intersecting one of the two lines at the point $(11, 7)$, and the other line at a point (x, y) . What is $x + y$? Express your answer as a common fraction.
- [A4]. A function $f(x)$ defined over the integers satisfies the following property:

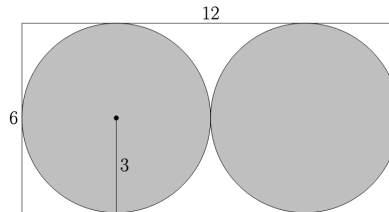
$$f(x + y) = 17^x f(y) + 17^y f(x)$$

Given that $f(2) = 2023$, find the last two digits of $f(6)$.

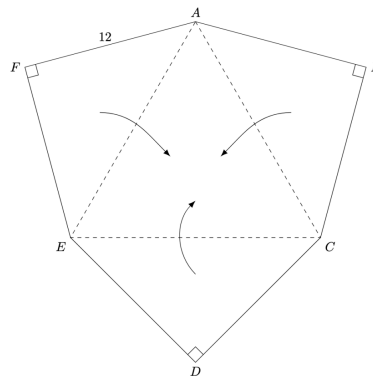
- [A5]. Let m be a real number. Suppose the equation $(x - 2023)\lfloor x \rfloor = m$ has k distinct real solutions in x that satisfy $x \geq 1$. What is the largest possible value of k ? Note: $\lfloor x \rfloor$ denotes the greatest integer that is less than or equal to x .

Geometry (G)

- [G1]. Square $ABCD$ has a point E on CD such that $AE = 13$ and $DE = 5$. What is BE^2 ?
- [G2]. John has a CD box that is 6 inches by 12 inches. He has two CDs of radius 3 inches, placed flat on the bottom without overlap. What is the radius, in inches, of the largest CD he can fit in the box without overlapping either of the first two CDs? Express your answer as a common fraction.



- [G3]. A piece of paper is in the shape of a convex hexagon $ABCDEF$, whose side lengths are all equal to 12. Additionally, $\angle ABC = \angle CDE = \angle EFA = 90^\circ$. The paper is folded along creases $\overline{AC}$, $\overline{CE}$, and $\overline{EA}$ such that vertices B , D , and F meet at the same point. The paper now forms the surface of a tetrahedron. What is the volume of this tetrahedron?



- [G4]. Let parallelogram $ABCD$ have $AB = 13$, $BC = 5$, and $AC < BD$. Let line $\overline{AD}$ and side $\overline{CD}$ intersect the circumcircle of $\triangle ABC$ at points $P \neq A$ and $Q \neq C$, respectively. Given that $\overline{PQ} \perp \overline{AB}$, find PQ . Express your answer as a common fraction.
- [G5]. In equilateral triangle ABC , let D be the midpoint of BC , and E the midpoint of AD . There exists point P on segment AB such that there is a point Q not on line AC satisfying $EQ = EP$, $\angle PEQ = 120^\circ$, and $QC \perp BC$. If $CP = 19$, what is $AP \cdot BP$? Express your answer as a common fraction.



Number Theory (N)

- [N1]. Let a and b be prime numbers such that $a + b = 50$. Find the greatest possible value of ab .
- [N2]. What is the largest power of 2 that can divide a 4-digit number of the form $\overline{ABAB}$ for nonzero digits A and B ?
- [N3]. Let a , b , and c be prime numbers such that $a^4 + b^4 + c^4 = 14738$. What is $a + b + c$?
- [N4]. Let a , b , and c be positive integers satisfying $\gcd(a, b) = 1296$, $\gcd(b, c) = 15$, and

$$\frac{\gcd(a, c)}{1296} > \left(\frac{\gcd(a, b)}{\gcd(a, c)}\right)^2.$$

How many possible values of c are there for $c < 1000$?

- [N5]. Let $1 = d_1 < d_2 < d_3 \dots < d_n = k$ be the distinct divisors of a positive integer $k > 1$. Call k special if the sequence of integers defined by $a_m = \gcd(d_m, d_{m+1})$ for all positive integers $m < n$ contains at least $n - 2$ distinct terms. Find the number of special numbers dividing $(202 \cdot 5)^6$.



Combinatorics (C)

- [C1]. Albert, Betty, and Carlos are 3 students in a class of 24 total students. The teacher randomly divides the class into 4 teams with 6 students in each. Given that Albert is not in the same team as Betty and Betty is not in the same team as Carlos, what is the probability that Carlos is in the same team as Albert? Express your answer as a common fraction.
- [C2]. How many distinct words can be made by a rearrangement of the word “MUSTMATH”, including the word “MUSTMATH” itself?
- [C3]. 9 points are evenly spaced around a circle, and 4 distinct points are chosen at random to form a quadrilateral. What is the probability that the quadrilateral contains the center of the circle? Express your answer as a common fraction.
- [C4]. What is the expected value of $1@2@3@4@5$, where each @ is either multiplication or division, with each operation being equally likely? Express your answer as a common fraction.
- [C5]. Let N be the number of ordered 4-tuples A, B, C, D of 4 not necessarily distinct non-empty subsets taken from the set $1, 2, 3, 4, 5, 6$ such that for each pair of subsets in the set, the sum of the elements in the intersection of the 2 subsets is odd. Then N can be expressed as $2^m \cdot n$ where m and n are integers and n isn't divisible by 2. Find $m + n$



Acceptable Answers

The following rules provide guidelines for acceptable answers in this round. Please note that any specifications provided in a problem will take precedence over these rules. The decisions of MMT coordinators are final.

- Common fractions are defined as a fraction in the form $\pm \frac{a}{b}$ where a and b are natural numbers and $\gcd(a, b) = 1$.
- Ratios and fractional answers should be expressed as common fractions unless otherwise specified.
- Radicals should be simplified. A simplified radical must satisfy:
 - No square factors, fractions, or nested radicals inside a radical
 - No radicals inside the denominator of a fraction
- Answers must be expressed to the exact accuracy called for in the problem (e.g. 25.0 will not be accepted for 25 and 25 will not be accepted for 25.0).
- Do not make approximations for numbers (e.g. 3.14 or $\frac{22}{7}$ for π) unless otherwise specified.
- Units **do not** need to be included but must be correct if included.